Random Forest

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Imports

In [41]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.ensemble import RandomForestClassifier as RFC
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report, ConfusionMatr
from sklearn import tree
from sklearn.model_selection import train_test_split , cross_val_score, Stratifi
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
from sklearn.metrics import roc_curve, auc
from sklearn.discriminant_analysis import StandardScaler
```

Helpfull Code

Load Data

In [42]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [66]:
```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported"])
x_copy = x.copy()
scaler = StandardScaler()
x = scaler.fit_transform(x)
```

Train-Test Split

In [44]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [45]:
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

בדיקת ערכי ההייפר פרמטרים

In [46]:
```
results = []
best_score = -1
best_params_rf = {}
```

n_estimators

In [47]:
```
results_n = []
best_score_n = 0
best_n = None

for n in range(50, 800, 50):

    rf = RFC(
        random_state=42,
        n_estimators=n,
        max_depth=None
    )

    fold_scores = cross_val_score(
        rf,
        x_train, y_train,
        cv=cv,
        scoring="accuracy"
    )

    mean_score = fold_scores.mean()

    results_n.append({
        "n_estimators": n,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })

    print(f"n_estimators: {n}, CV Accuracy: {100*mean_score:.2f}%")
    if mean_score > best_score_n:
        best_score_n = mean_score
        best_n = n

print("Best n_estimators:", best_n)
print(f"Best CV accuracy: {100*best_score_n:.2f}%")

results_n_df = pd.DataFrame(results_n).sort_values("cv_mean", ascending=False)
display(results_n_df.head(10).round(4))
```

```
n_estimators: 50, CV Accuracy: 79.77%
n_estimators: 100, CV Accuracy: 79.83%
n_estimators: 150, CV Accuracy: 79.71%
n_estimators: 200, CV Accuracy: 79.80%
n_estimators: 250, CV Accuracy: 79.96%
n_estimators: 300, CV Accuracy: 79.92%
n_estimators: 350, CV Accuracy: 79.83%
n_estimators: 400, CV Accuracy: 79.86%
n_estimators: 450, CV Accuracy: 79.85%
n_estimators: 500, CV Accuracy: 79.86%
n_estimators: 550, CV Accuracy: 79.82%
n_estimators: 600, CV Accuracy: 79.82%
n_estimators: 650, CV Accuracy: 79.71%
n_estimators: 700, CV Accuracy: 79.67%
n_estimators: 750, CV Accuracy: 79.70%
Best n_estimators: 250
Best CV accuracy: 79.96%
```

	n_estimators	cv_mean	cv_std
4	250	0.7996	0.0067
5	300	0.7992	0.0069
7	400	0.7986	0.0086
9	500	0.7986	0.0085
8	450	0.7985	0.0078
6	350	0.7983	0.0075
1	100	0.7983	0.0066
11	600	0.7982	0.0084
10	550	0.7982	0.0078
3	200	0.7980	0.0070

max_depth

In [48]:
```
results_depth = []
best_score_depth = 0
best_depth = None

for depth in range(5, 25):

    rf = RFC(
        random_state=42,
        n_estimators=best_n,
        max_depth=depth
    )

    fold_scores = cross_val_score(
        rf,
        x_train, y_train,
        cv=cv,
        scoring="accuracy"
    )
```

```
    mean_score = fold_scores.mean()

    results_depth.append({
        "max_depth": depth,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })

    if mean_score > best_score_depth:
        best_score_depth = mean_score
        best_depth = depth

print("Best max_depth:", best_depth)
print(f"Best CV accuracy: {100*best_score_depth:.2f}%")

results_depth_df = pd.DataFrame(results_depth).sort_values("cv_mean", ascending=
display(results_depth_df.head(10).round(4))
```

```
Best max_depth: 21
Best CV accuracy: 79.87%
```

	max_depth	cv_mean	cv_std
16	21	0.7987	0.0069
17	22	0.7980	0.0071
10	15	0.7979	0.0038
18	23	0.7974	0.0090
19	24	0.7974	0.0081
8	13	0.7967	0.0043
11	16	0.7967	0.0019
12	17	0.7967	0.0042
9	14	0.7964	0.0047
15	20	0.7964	0.0067

max_features

In [49]:
```
results_feat = []
best_score_final = 0
best_params_rf = None

for feat in ["sqrt", "log2", None]:

    rf = RFC(
        random_state=42,
        n_estimators=best_n,
        max_depth=best_depth,
        max_features=feat
    )

    fold_scores = cross_val_score(
        rf,
        x_train, y_train,
        cv=cv,
```

```
        scoring="accuracy"
    )

    mean_score = fold_scores.mean()

    results_feat.append({
        "max_features": feat,
        "cv_mean": mean_score,
        "cv_std": fold_scores.std()
    })
    if mean_score > best_score_final:
        best_score_final = mean_score
        best_params_rf = {
            "n_estimators": best_n,
            "max_depth": best_depth,
            "max_features": feat
        }

print("\nBest RF params:", best_params_rf)
print(f"Best CV accuracy: {100*best_score_final:.2f}%")

results_feat_df = pd.DataFrame(results_feat).sort_values("cv_mean", ascending=Fa
display(results_feat_df.round(4))
```

```
Best RF params: {'n_estimators': 250, 'max_depth': 21, 'max_features': 'sqrt'}
Best CV accuracy: 79.87%
```

	max_features	cv_mean	cv_std
0	sqrt	0.7987	0.0069
1	log2	0.7987	0.0069
2	None	0.7882	0.0061

min_sample_leaf

In [50]:

```
results_leaf = []

best_score = 0
best_leaf = None

for leaf in range(5, 31, 1):

    rf = RFC(
        random_state=42,
        n_estimators=best_params_rf["n_estimators"],
        max_depth=best_params_rf["max_depth"],
        max_features=best_params_rf["max_features"],
        min_samples_leaf=leaf
    )

    scores = cross_val_score(
        rf,
        x_train, y_train,
        cv=cv,
        scoring="accuracy"
    )

    mean_score = scores.mean()
```

```
    results_leaf.append({
        "min_samples_leaf": leaf,
        "cv_mean": mean_score,
        "cv_std": scores.std()
    })

    if mean_score > best_score:
        best_score = mean_score
        best_leaf = leaf

print("Best min_samples_leaf:", best_leaf)
print(f"Best CV accuracy: {100*best_score:.2f}%")

display(pd.DataFrame(results_leaf).sort_values("cv_mean", ascending=False).round
```

```
Best min_samples_leaf: 7
Best CV accuracy: 79.76%
```

	min_samples_leaf	cv_mean	cv_std
2	7	0.7976	0.0043
3	8	0.7974	0.0072
0	5	0.7963	0.0047
1	6	0.7961	0.0062
4	9	0.7952	0.0059
5	10	0.7951	0.0067
8	13	0.7941	0.0068
6	11	0.7941	0.0063
14	19	0.7933	0.0098
11	16	0.7933	0.0079
9	14	0.7932	0.0093
17	22	0.7930	0.0091
13	18	0.7930	0.0099
12	17	0.7928	0.0087
16	21	0.7922	0.0095
7	12	0.7920	0.0064
19	24	0.7919	0.0097
10	15	0.7919	0.0084
24	29	0.7917	0.0107
15	20	0.7910	0.0095
21	26	0.7909	0.0088
20	25	0.7907	0.0094
23	28	0.7901	0.0121
18	23	0.7901	0.0100
22	27	0.7900	0.0109
25	30	0.7895	0.0108

run best Random Forest

In [77]:
```
# יצירת טבלת הייפר-פרמטרים
rf_params_table = pd.DataFrame({
    "Hyperparameter": [
        "n_estimators",
        "max_depth",
        "max_features",
        "min_samples_leaf"
```

```
    ],
    "Best Value": [
        best_params_rf["n_estimators"],
        best_params_rf["max_depth"],
        best_params_rf["max_features"],
        best_leaf
    ]
})

display(rf_params_table)
```

	Hyperparameter	Best Value
0	n_estimators	250
1	max_depth	21
2	max_features	sqrt
3	min_samples_leaf	7

In [51]:
```
best_RF = RFC(
    random_state=42,
    n_estimators=best_params_rf["n_estimators"],
    max_depth=best_params_rf["max_depth"],
    max_features=best_params_rf["max_features"],
    min_samples_leaf=best_leaf

)
best_RF.fit(x_train, y_train)
```

Out[51]:
```
RandomForestClassifier
RandomForestClassifier(max_depth=21, min_samples_leaf=7, n_estimators=2
50,
                       random_state=42)
```

In [52]:
```
best_RF.get_params()
```

Out[52]:
```
{'bootstrap': True,
 'ccp_alpha': 0.0,
 'class_weight': None,
 'criterion': 'gini',
 'max_depth': 21,
 'max_features': 'sqrt',
 'max_leaf_nodes': None,
 'max_samples': None,
 'min_impurity_decrease': 0.0,
 'min_samples_leaf': 7,
 'min_samples_split': 2,
 'min_weight_fraction_leaf': 0.0,
 'monotonic_cst': None,
 'n_estimators': 250,
 'n_jobs': None,
 'oob_score': False,
 'random_state': 42,
 'verbose': 0,
 'warm_start': False}
```

Train Acuuracy

In [53]:
```
y_pred_train  = best_RF.predict(x_train)
train_acc = accuracy_score(y_train, y_pred_train)
```

In [54]:
```
print("="*40)
print("BEST Random Forest — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc*100:.2f}%")
print("="*40)
```

```
========================================
BEST Random Forest — TRAIN SET PERFORMANCE
========================================
Accuracy : 87.62%
========================================
```

Results

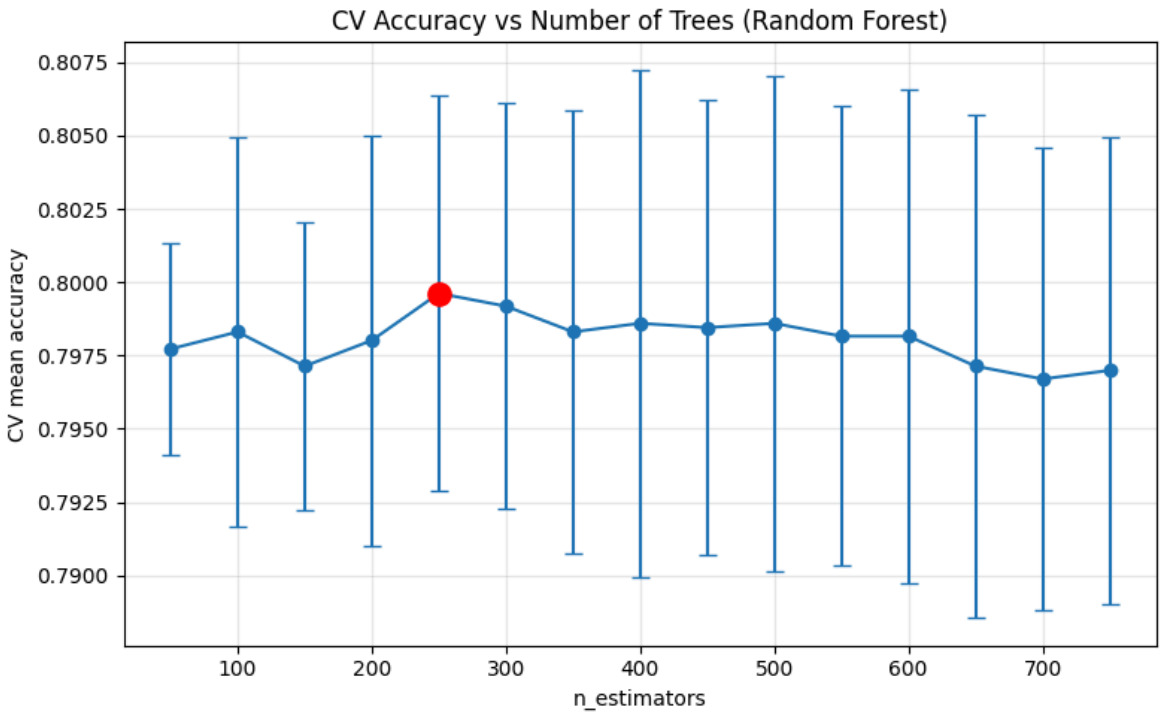
In [55]:
```
y_pred_train = best_RF.predict(x_train)
```

In [56]:
```
results_n = pd.DataFrame(results_n)
results_depth = pd.DataFrame(results_depth)
results_feat = pd.DataFrame(results_feat)
results_leaf = pd.DataFrame(results_leaf)
```

best n_estimator accuracy

In [57]:
```
plt.figure(figsize=(8,5))

plt.errorbar(
    results_n["n_estimators"],
    results_n["cv_mean"],
    yerr=results_n["cv_std"],
    fmt="o-",
    capsize=4
)
```

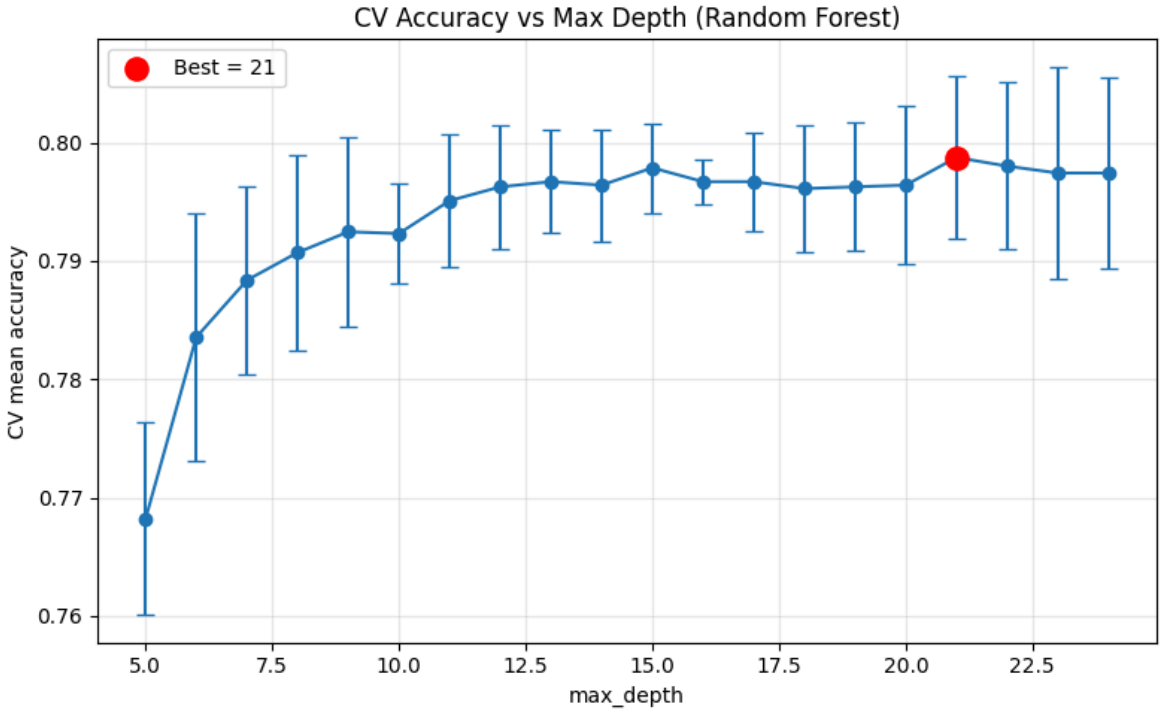
```
best_idx = results_n["cv_mean"].idxmax()
best_n = results_n.loc[best_idx, "n_estimators"]
best_score = results_n.loc[best_idx, "cv_mean"]

plt.scatter(
    best_n,
    best_score,
    color="red",
    s=120,
    zorder=5,
    label=f"Best = {best_n}"
)


plt.title("CV Accuracy vs Number of Trees (Random Forest)")
plt.xlabel("n_estimators")
plt.ylabel("CV mean accuracy")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



best depth accuracy

In [58]:
```
plt.figure(figsize=(8,5))

plt.errorbar(
    results_depth["max_depth"],
    results_depth["cv_mean"],
    yerr=results_depth["cv_std"],
    fmt="o-",
    capsize=4
)

best_idx = results_depth["cv_mean"].idxmax()
best_depth = results_depth.loc[best_idx, "max_depth"]
best_score = results_depth.loc[best_idx, "cv_mean"]
```

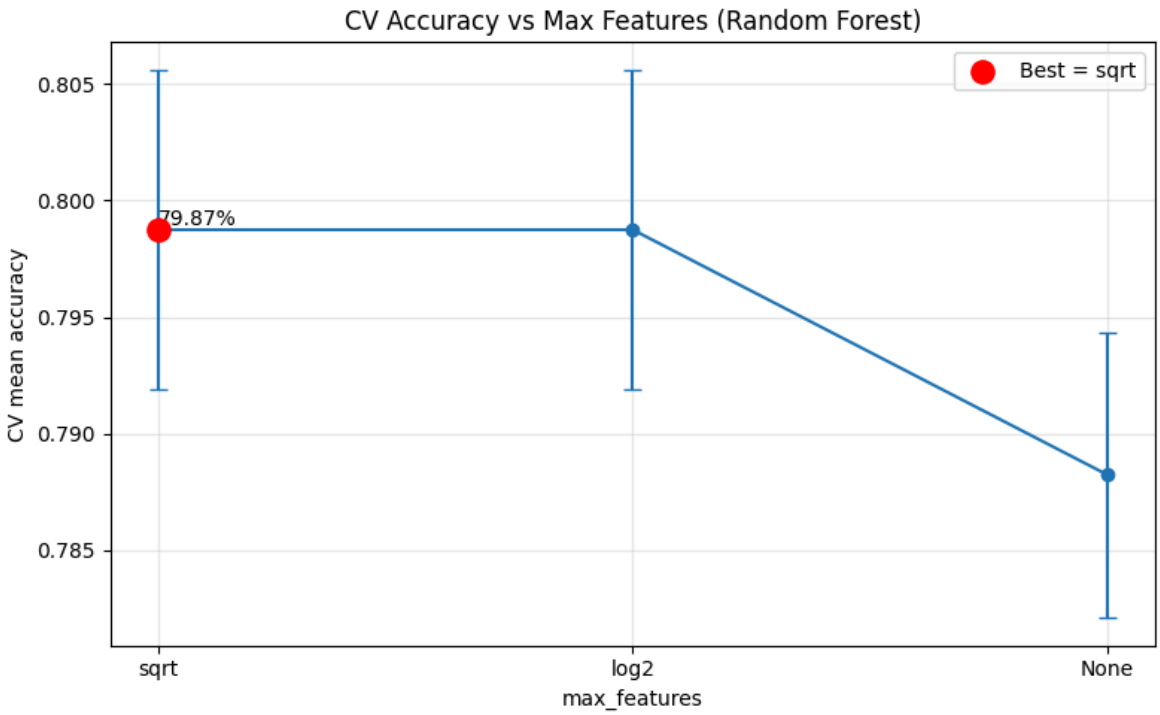
```
plt.scatter(
    best_depth,
    best_score,
    color="red",
    s=120,
    zorder=5,
    label=f"Best = {best_depth}"
)

plt.title("CV Accuracy vs Max Depth (Random Forest)")
plt.xlabel("max_depth")
plt.ylabel("CV mean accuracy")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



best max_features accuracy

In [59]:
```
results_feat["max_features"] = results_feat["max_features"].astype(str)

plt.figure(figsize=(8,5))

# גרף
plt.errorbar(
    results_feat["max_features"],
    results_feat["cv_mean"],
    yerr=results_feat["cv_std"],
    fmt="o-",
    capsize=4
)

# נקודה הכי טובה
best_idx = results_feat["cv_mean"].idxmax()
best_feat = results_feat.loc[best_idx, "max_features"]
best_score = results_feat.loc[best_idx, "cv_mean"]
```

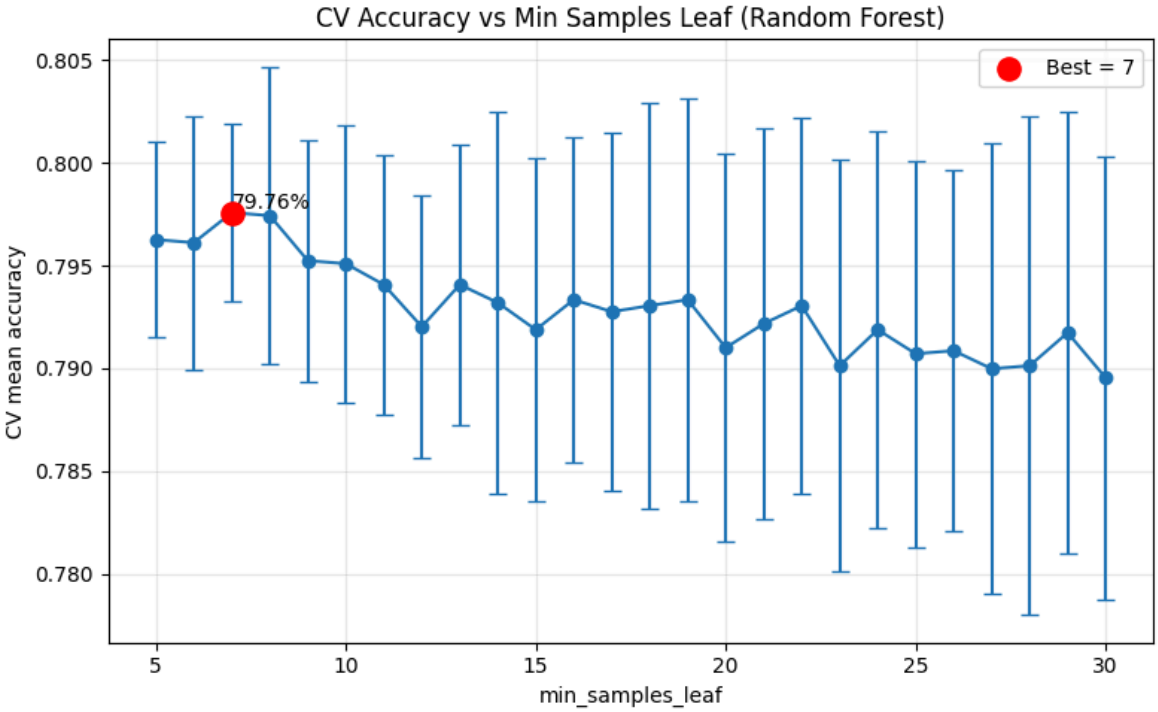
```
plt.scatter(
    best_feat,
    best_score,
    color="red",
    s=120,
    zorder=5,
    label=f"Best = {best_feat}"
)

# אחוז ליד הנקודה
plt.text(
    best_feat,
    best_score,
    f"{100*best_score:.2f}%",
    ha="left",
    va="bottom"
)

plt.title("CV Accuracy vs Max Features (Random Forest)")
plt.xlabel("max_features")
plt.ylabel("CV mean accuracy")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



best min_samples_leaf accuracy

In [60]:
```
plt.figure(figsize=(8,5))

# גרף
plt.errorbar(
    results_leaf["min_samples_leaf"],
    results_leaf["cv_mean"],
    yerr=results_leaf["cv_std"],
    fmt="o-",
    capsize=4
```

```
)

# הנקודה הטובה ביותר
best_idx = results_leaf["cv_mean"].idxmax()
best_leaf = results_leaf.loc[best_idx, "min_samples_leaf"]
best_score = results_leaf.loc[best_idx, "cv_mean"]

plt.scatter(
    best_leaf,
    best_score,
    color="red",
    s=120,
    zorder=5,
    label=f"Best = {best_leaf}"
)

# אחוז ליד הנקודה
plt.text(
    best_leaf,
    best_score,
    f"{100*best_score:.2f}%",
    ha="left",
    va="bottom"
)

plt.title("CV Accuracy vs Min Samples Leaf (Random Forest)")
plt.xlabel("min_samples_leaf")
plt.ylabel("CV mean accuracy")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



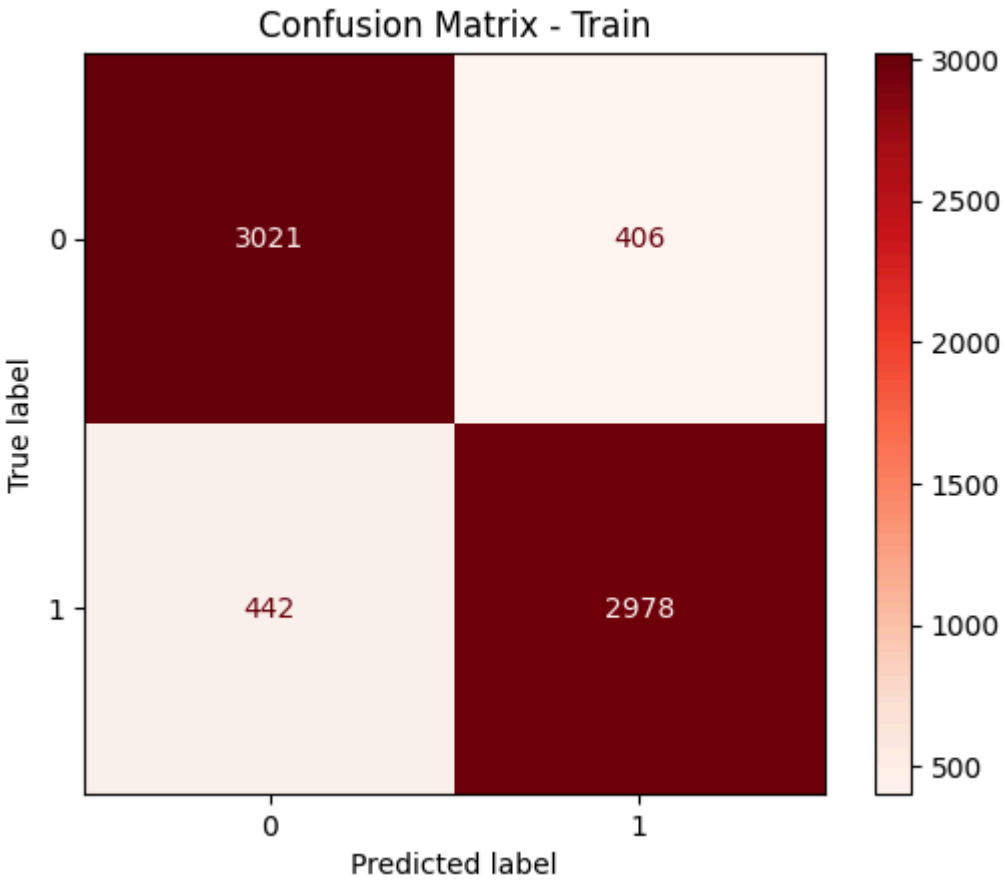
best Random Forset accuracy

In [61]:
```
accuracy_train = best_RF.score(x_train, y_train)
print(f'Accuracy Train: {accuracy_train}')
```

```
Accuracy Train: 0.8761501387468964
```

Confusion matrix

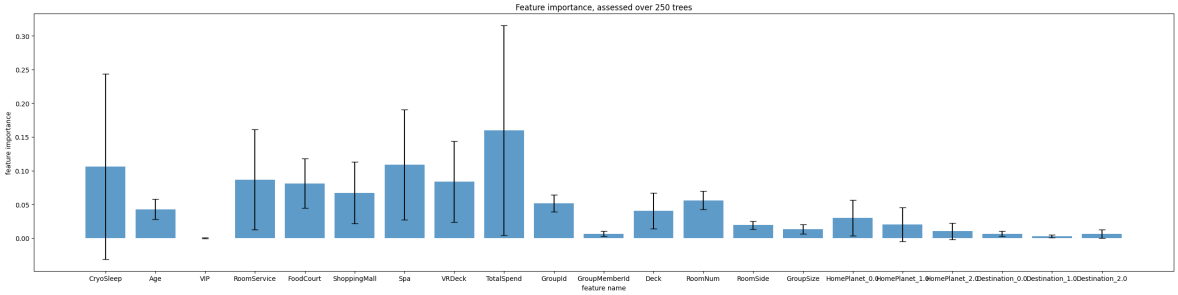
In [62]:
```
cm_train = confusion_matrix(y_train, best_RF.predict(x_train))

disp_train = ConfusionMatrixDisplay(confusion_matrix=cm_train)
disp_train.plot(cmap='Reds')
plt.title('Confusion Matrix - Train')
plt.show()
```



Random Forest Feature Importance

In [69]:
```
feature_importances = np.array([tree.feature_importances_ for tree in best_RF.es

feature_names = x_copy.columns.tolist()
mean_importance = feature_importances.mean(axis=0)
std_importance = feature_importances.std(axis=0)

positions = np.arange(feature_importances.shape[1])


plt.figure(figsize=(25, 6))
plt.bar(
    positions,
    mean_importance,
    yerr=std_importance,
    capsize=5,
    alpha=0.7
)
```

```
plt.xticks(positions, feature_names, rotation=0, ha="center")
plt.tight_layout()

plt.xlabel("feature name")
plt.ylabel("feature importance")
plt.title(f"Feature importance, assessed over {feature_importances.shape[0]} tre
plt.show()
```



In [71]:
```
fi_per_tree = np.array([t.feature_importances_ for t in best_RF.estimators_])

fi_mean = fi_per_tree.mean(axis=0)
fi_std  = fi_per_tree.std(axis=0)

fi_df = pd.DataFrame({
    "feature": x_copy.columns,
    "importance_mean": fi_mean,
    "importance_std": fi_std
}).sort_values("importance_mean", ascending=False)

print("=== Feature Importances (Best Forest) ===")
display(fi_df)
```

```
=== Feature Importances (Best Forest) ===
```

	feature	importance_mean	importance_std
8	TotalSpend	0.159675	0.155606
6	Spa	0.108774	0.081476
0	CryoSleep	0.106330	0.137574
3	RoomService	0.086714	0.074475
7	VRDeck	0.083720	0.059717
4	FoodCourt	0.081182	0.036544
5	ShoppingMall	0.067302	0.045848
12	RoomNum	0.056153	0.013744
9	GroupId	0.051665	0.012372
1	Age	0.042697	0.015074
11	Deck	0.040302	0.026433
15	HomePlanet_0.0	0.029806	0.026564
16	HomePlanet_1.0	0.020239	0.025184
13	RoomSide	0.019232	0.006127
14	GroupSize	0.013436	0.006835
17	HomePlanet_2.0	0.010370	0.012121
18	Destination_0.0	0.006565	0.003713
10	GroupMemberId	0.006545	0.003632
20	Destination_2.0	0.006428	0.006129
19	Destination_1.0	0.002560	0.002046
2	VIP	0.000305	0.000704

Summary Table

In [72]:
```
summary_df = pd.DataFrame([{
    "best_n_estimators": best_params_rf["n_estimators"],
    "best_max_depth": best_params_rf["max_depth"],
    "n_features": x_train.shape[1],
    "n_samples": x_train.shape[0]
}])

print("=== Best Forest Summary ===")
display(summary_df)
```

```
=== Best Forest Summary ===
```

	best_n_estimators	best_max_depth	n_features	n_samples
0	250	21	21	6847

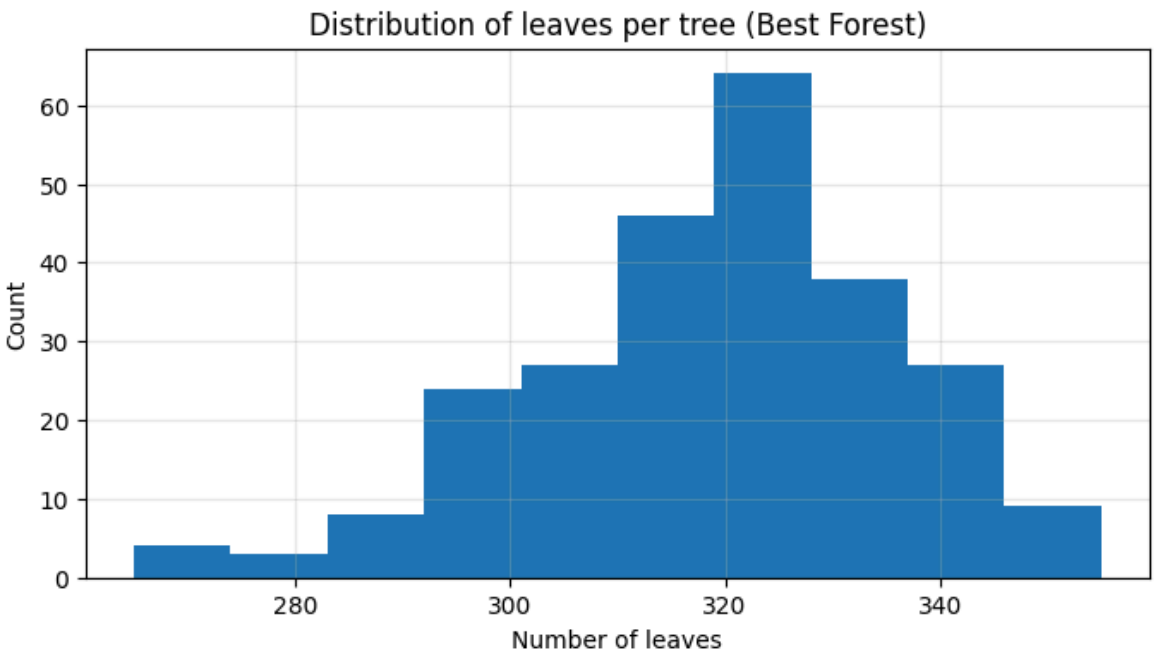
Statistics

In [73]:
```
tree_depths = np.array([t.tree_.max_depth for t in best_RF.estimators_])
tree_nodes  = np.array([t.tree_.node_count for t in best_RF.estimators_])
tree_leaves = np.array([t.tree_.n_leaves for t in best_RF.estimators_])

stats_df = pd.DataFrame({
    "metric": ["depth", "node_count", "n_leaves"],
    "mean":   [tree_depths.mean(), tree_nodes.mean(), tree_leaves.mean()],
    "std":    [tree_depths.std(),  tree_nodes.std(),  tree_leaves.std()],
    "min":    [tree_depths.min(),  tree_nodes.min(),  tree_leaves.min()],
    "max":    [tree_depths.max(),  tree_nodes.max(),  tree_leaves.max()],
})

print("=== Trees Statistics (actual) ===")
display(stats_df.round(2))
```

```
=== Trees Statistics (actual) ===
```

	metric	mean	std	min	max
0	depth	17.81	1.62	14	21
1	node_count	636.32	33.63	529	709
2	n_leaves	318.66	16.81	265	355

In [74]:
```
plt.figure(figsize=(8,4))
plt.hist(tree_leaves, bins=10)
plt.xlabel("Number of leaves")
plt.ylabel("Count")
plt.title("Distribution of leaves per tree (Best Forest)")
plt.grid(alpha=0.3)
plt.show()
```



In [75]:
```
acc = accuracy_score(y_train, y_pred_train)
prec = precision_score(y_train, y_pred_train)
rec = recall_score(y_train, y_pred_train)
f1 = f1_score(y_train, y_pred_train)
```

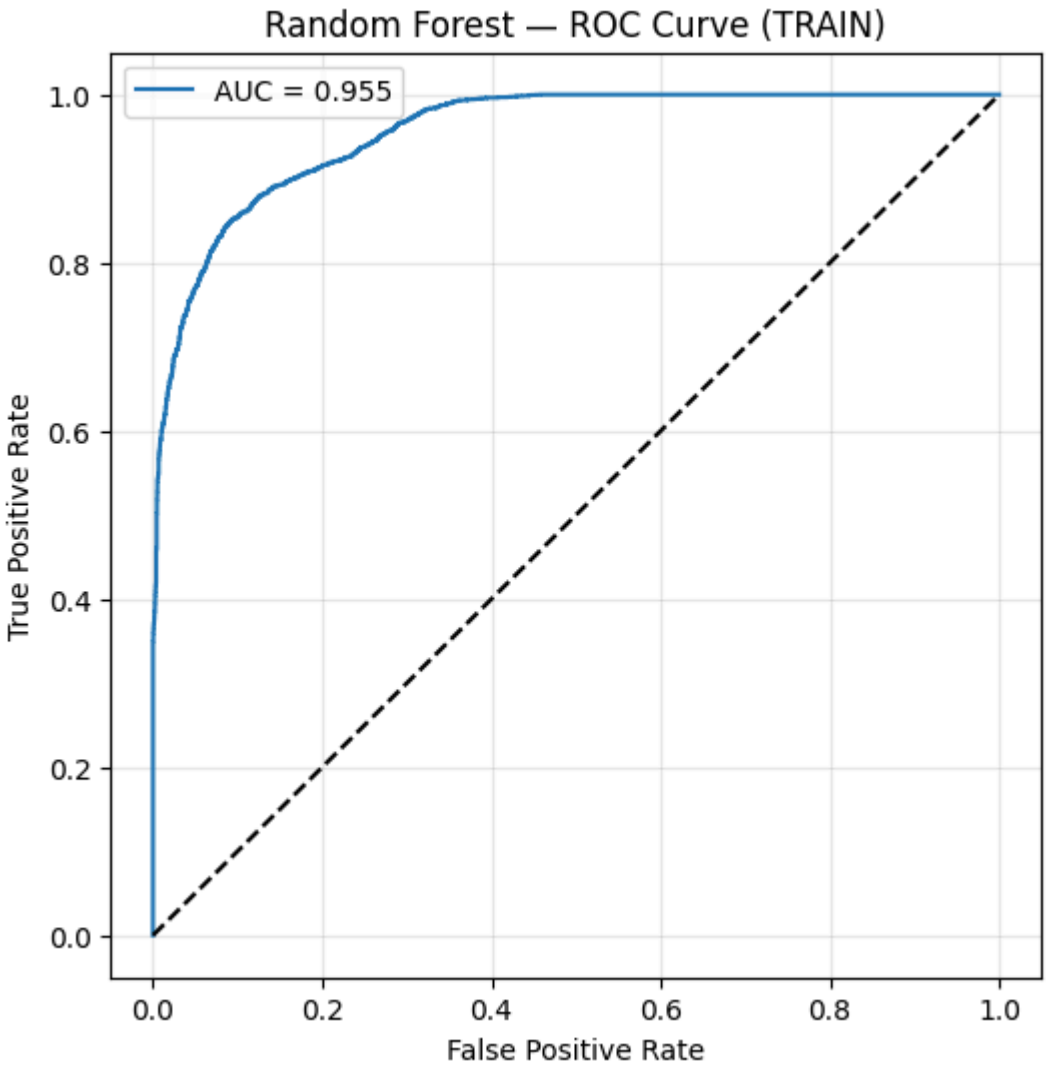
```
metrics_df = pd.DataFrame([{
    "model": "RandomForest",
    "accuracy": acc,
    "precision": prec,
    "recall": rec,
    "f1_score": f1
}])

display(metrics_df.round(4))

print("\nClassification Report (TRAIN):")
print(classification_report(y_train, y_pred_train))
```

	model	accuracy	precision	recall	f1_score
0	RandomForest	0.8762	0.88	0.8708	0.8754

```
Classification Report (TRAIN):
              precision    recall  f1-score   support

           0       0.87      0.88      0.88      3427
           1       0.88      0.87      0.88      3420

    accuracy                           0.88      6847
   macro avg       0.88      0.88      0.88      6847
weighted avg       0.88      0.88      0.88      6847
```

In [76]:

```
y_prob_train = best_RF.predict_proba(x_train)[:, 1]

fpr, tpr, thresholds = roc_curve(y_train, y_prob_train)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,6))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], "k--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Random Forest – ROC Curve (TRAIN)")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```


Random Forest — ROC Curve (TRAIN)
AUC = 0.955
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0